

Arash Jalil Khabbazi, MS

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Profile

I work on reinforcement learning (RL) and model predictive control (MPC) for smart buildings and energy systems. I also explore broader machine learning (ML) methods in other AI-enabled applications

Education

Purdue University, IN, USA PhD in Mechanical Engineering — GPA: 3.86/4.0 Minor in Computational Science and Engineering	2023 – 2027
University of British Columbia, BC, Canada MS in Mechanical Engineering — GPA: 4.0/4.0 (94%)	2021 – 2023
University of Tabriz, EA, Iran BS in Mechanical Engineering — GPA: 4.0/4.0 (19.12/20) — Rank: 1/155+	2016 – 2020

Experience

Applied ML Intern, Tesla, Inc., Sparks, NV, USA 05/2025 – 08/2025

- Built supervised ML models (XGBoost, neural nets, SVR, and ensembles) for defect detection and quality assessment.
- Contributed to deployment of a thermal imaging system for non-destructive inspection in drive-unit manufacturing.
- Developed data pipelines in MySQL and engineered features to extract high-signal inputs for ML models.
- Collaborated on projects projected to reduce testing costs by over \$1M annually through improved quality assessment.

PhD Research Assistant, Purdue University, West Lafayette, IN, USA 08/2023 – 08/2027

- Developed a Soft Actor-Critic RL HVAC controller in [OCHRE-Gym](#), saving 17.5% energy while maintaining comfort.
- Supported commissioning of [Herrick Labs](#) by upgrading the building automation system and integrating Modbus.
- Created real-time dashboards for energy/comfort and ML models for occupancy detection (98% accuracy).
- Designed HVAC control strategies with MPC and DeePC, showing DeePC matched MPC without a system model.
- Reviewed 100+ publications on MPC and RL in HVAC, leading to a published [review paper](#) in Applied Energy.
- Assisted in developing active current-limiting control for appliances to prevent breaker overloads in electrified homes.

Student Researcher, FortisBC Energy Inc., Surrey, BC, Canada 05/2021 – 08/2023

- Performed assessments on hydrogen injection into natural gas pipelines and renewable gas projects (UBC H₂Lab).
- Worked with engineers and research teams (materials, sensors, AI, safety) to assess industry impacts.

MS Research Assistant, University of British Columbia, BC, Canada 05/2021 – 08/2023

- Modeled hydrogen injection into natural gas pipelines with CFD by developing ideal- and real-gas mixture models.
- Produced a [journal paper](#), three conference papers, two awards, and a [master's thesis](#) based on the research findings.

Skills

- Programming:** Python, C, MATLAB, R, MySQL, Git, Linux
- Machine Learning:** PyTorch, TensorFlow, scikit-learn, Stable-Baselines3, XGBoost
- Data & Analytics:** NumPy, Pandas, SciPy, Matplotlib, Seaborn, Grafana
- Modeling & Optimization:** CVX, CVXPY
- Simulation & Control Testbeds:** BOPTEST, EnergyPlus, OpenAI Gym, OCHRE, OCHRE-Gym, TRNSYS
- Systems & Integration:** Tridium Niagara, InfluxDB, Modbus, IO-Link, MTConnect

Selected Publications — Full list on [Google Scholar](#)

- A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premier, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, K.J. Kircher. “Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review.” *Applied Energy*, 2025. [\[DOI\]](#)
- A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, J. Quinn. “Mixing hydrogen into natural gas distribution pipeline system through Tee junctions.” *International Journal of Hydrogen Energy*, 2024. [\[DOI\]](#)
- Six peer-reviewed conference papers (ACM BuildSys, CISBAT, IHPB, CSME, IGEC; 2021–2025).

Selected Honors & Awards

Best PhD Forum Presentation Runner-up Award at ACM BuildSys (2025) — ASHRAE Graduate Student Grant-in-Aid Award (2025) — Best Paper Award at CSME (2023) — Best Presentation Award at CSME (2022) — UBC Graduate Scholarship (2022) — UBC Dean’s Entrance Scholarship (2021)

Selected Courses

Artificial Intelligence — Applied Machine Learning — Reinforcement Learning and Control — Intro to Convex Optimization — Industrial IoT Implementation — Statistical Methods — Distributed Energy Resources — Analysis of Thermal Systems — Advanced Thermodynamics